

JAVA – CO2AT1

Q1.

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JAVA PROGRAMMING

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CO2-AT1-Assignment1

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QUESTIONS

Q1
Smart Document Editor using
String, StringBuilder and
StringBuffer
10 marks

Q2
Hospital Token Management
System using List Interface
10 marks

Q3
Airport Boarding Queue
Management using Queue
Interface
10 marks

Q4
Digital Certificate Verification
using Set Interface
10 marks

Q1 Smart Document Editor using String, StringBuilder and StringBuffer 10 marks

Question 1 :

- 1.Read a paragraph from the user.
- 2.Count the number of characters, words, and sentences.
- 3.Replace every occurrence of a given word with another word.
- 4.Reverse each sentence individually.
- 5.Maintain editing history using StringBuffer.
- 6.Construct the final formatted document using StringBuilder.
- 7.Display the original document, editing history, and final document.

Requirement: Use String methods wherever appropriate. Use StringBuilder for document construction and StringBuffer for edit history.

Your Code

```
1 import java.util.Scanner;
2 public class SmartDocumentEditor {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         StringBuffer editHistory = new StringBuffer();
```

Input

Java is fun. Java is powerful.
Programming is fun.
java
Python

using Set Interface
10 marks

Q5
Smart Inventory Management
using Map Interface
10 marks

5/5 answered

```
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         StringBuffer editHistory = new StringBuffer();
6         System.out.println("Enter a paragraph:");
7         String paragraph = sc.nextLine();
8         String originalDocument = paragraph;
9         editHistory.append("Original Document Created.\n");
10        int charCount = paragraph.replaceAll("\\s", "").length();
11        int wordCount = paragraph.trim().split("\\s+").length;
12        String[] sentences = paragraph.split("[.!?\\s]*");
13        int sentenceCount = 0;
14        for (String s : sentences) {
15            if (!s.trim().isEmpty()) {
16                sentenceCount++;
17            }
18        }
19        editHistory.append("Counted Characters: ").append(charCount)
20            .append(", Words: ").append(wordCount)
21            .append(", Sentences: ").append(sentenceCount).append("\n");
22        System.out.println("Enter the word to replace:");
23        String oldWord = sc.nextLine();
24        System.out.println("Enter the new word:");
25        String newWord = sc.nextLine();
26
27        String replacedText = paragraph.replaceAll("\\b" + oldWord +
28        editHistory.append("Replaced ").append(oldWord)
```

java
Python

Output

Enter a paragraph:
Enter the word to replace:
Enter the new word:
==== ORIGINAL DOCUMENT ====
Java is fun. Java is powerful.

```

34         if (!sentence.trim().isEmpty()) {
35             StringBuilder sb = new StringBuilder(sentence.trim());
36             sb.reverse();
37             reversedDocument.append(sb).append(" ");
38         }
39     }
40
41     editHistory.append("Reversed each sentence individually.\n");
42     StringBuilder finalDocument = new StringBuilder();
43     finalDocument.append("----- FINAL DOCUMENT ----- \n");
44     finalDocument.append(reversedDocument.toString().trim());
45     finalDocument.append("\n-----");
46     editHistory.append("Final document constructed successfully.");
47     System.out.println("\n===== ORIGINAL DOCUMENT =====");
48     System.out.println(originalDocument);
49     System.out.println("\n===== EDIT HISTORY =====");
50     System.out.println(editHistory.toString());
51     System.out.println("\n===== FINAL DOCUMENT =====");
52     System.out.println(finalDocument.toString());
53     sc.close();
54 }
55 }

```

Run

Save

Q2.

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QUESTIONS

Q1

Smart Document Editor using String, StringBuilder and StringBuffer

10 marks

Q2

Hospital Token Management System using List Interface

10 marks

Q3

Airport Boarding Queue Management using Queue Interface

10 marks

Q4

Digital Certificate Verification using Set Interface

10 marks

Q5

Smart Inventory Management using Map Interface

10 marks

5/5 answered

Q2 Hospital Token Management System using List Interface

10 marks

Hospital Token Management System using List Interface

Store patient details using an appropriate List implementation.

Register new patients.

Display patients in registration order.

Update patient information using Patient ID.

Remove cancelled appointments.

Display all patients of a particular department.

Find the first and last registered patients.

Requirement: Use a suitable List implementation

Your Code

```
1 import java.util.*;
2 public class Main {
```

Input

3
101

Your Code

```
1 import java.util.*;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         ArrayList<Patient> list = new ArrayList<>();
6         int n = sc.nextInt();
7         sc.nextLine();
8         for (int i = 0; i < n; i++) {
9             int id = sc.nextInt();
10            sc.nextLine();
11            String name = sc.nextLine();
12            String dept = sc.nextLine();
13            list.add(new Patient(id, name, dept));
14        }
15        System.out.println("All Patients:");
16        for (Patient p : list) {
17            System.out.println(p.id + " " + p.name + " " + p.dept);
18        }
19        int id = sc.nextInt();
20        System.out.println("The Entered patien ID to update is : "+
```

Input

3
101
Ram
Cardiology

Output

All Patients:
101 Ram Cardiology
102 Ravi Neurology
103 Kumar Cardiology
The Entered patien ID to update is : 102

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```
27     }
28 }
29 String dep = sc.nextLine();
30 System.out.println("Enter Department to search : "+dep);
31 System.out.println("Patients in " + dep + " Department:");
32 for (Patient p : list) {
33     if (p.dept.equalsIgnoreCase(dep)) {
34         System.out.println(p.id + " " + p.name);
35     }
36 }
37 System.out.println("\nFirst Registered Patient:");
38 System.out.println(list.get(0).id + " " + list.get(0).name);
39 System.out.println("Last Registered Patient:");
40 System.out.println(list.get(list.size() - 1).id + " " + list
41 )
42 }
43 class Patient {
44     int id;
45     String name;
46     String dept;
47     Patient(int id, String name, String dept) {
48         this.id = id;
49         this.name = name;
50         this.dept = dept;
51     }
52 }
```

Q3.



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QUESTIONS
Q1
Smart Document Editor using String, StringBuilder and StringBuffer
10 marks
Q2
Hospital Token Management System using List Interface
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Airport Boarding Queue Management using Queue Interface
10 marks
Q4
Digital Certificate Verification using Set Interface
10 marks
Q5
Smart Inventory Management using Map Interface
10 marks

Q3 Airport Boarding Queue Management using Queue Interface
10 marks

Airport Boarding Queue Management using Queue Interface

Add passengers to the boarding queue.

Display the current queue.

Board passengers in FIFO order.

Display the next passenger without removing them.

Handle VIP passengers using a priority-based queue.

Display remaining passengers after each boarding operation.

Requirement: Use appropriate Queue implementations.

Your Code

```
1 import java.util.*;
2 public class Employee {
3     public static void main(String[] args) {
```

Input

```
3
Arun
Kumar
```



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using Set Interface
10 marks
Q5
Smart Inventory Management using Map Interface
10 marks
5/5 answered

```
1 import java.util.*;
2 public class Employee {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         Queue<String> boardingQueue = new LinkedList<>();
6         PriorityQueue<String> vipQueue = new PriorityQueue<>();
7         int n = sc.nextInt();
8         sc.nextLine();
9         for (int i = 0; i < n; i++) {
10             boardingQueue.add(sc.nextLine());
11         }
12         int m = sc.nextInt();
13         sc.nextLine();
14         for (int i = 0; i < m; i++) {
15             vipQueue.add(sc.nextLine());
16         }
17         System.out.println("Current Boarding Queue:");
18         for (String p : boardingQueue) {
19             System.out.println(p);
20         }
21         if (!boardingQueue.isEmpty()) {
22             System.out.println("Next Passenger: " + boardingQueue.poll());
23         }
24         while (!boardingQueue.isEmpty()) {
25             System.out.println("Boarded: " + boardingQueue.poll());
26             System.out.println("Remaining Queue:");
27             for (String p : boardingQueue) {
28                 System.out.println(p);
29             }
30         }
31     }
32 }
```

Input

```
3
Arun
Kumar
Rahul
```

Output

```
Current Boarding Queue:
Arun
Kumar
Rahul
Next Passenger: Arun
Boarded: Arun
```

```
20     }
21     if (!boardingQueue.isEmpty()) {
22         System.out.println("Next Passenger: " + boardingQueue.poll());
23     }
24     while (!boardingQueue.isEmpty()) {
25         System.out.println("Boarded: " + boardingQueue.poll());
26         System.out.println("Remaining Queue:");
27         if (boardingQueue.isEmpty()) {
28             System.out.println("Empty");
29         } else {
30             for (String p : boardingQueue) {
31                 System.out.println(p);
32             }
33         }
34     }
35     System.out.println("VIP Boarding Order:");
36     while (!vipQueue.isEmpty()) {
37         System.out.println(vipQueue.poll());
38     }
39     sc.close();
40 }
41 }
```

Run

Save

Q4.

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QUESTIONS**Q1** ✓
Smart Document Editor using String, StringBuilder and StringBuffer
10 marks**Q2** ✓
Hospital Token Management System using List Interface
10 marks**Q3** ✓
Airport Boarding Queue Management using Queue Interface
10 marks**Q4** ✓
Digital Certificate Verification using Set Interface
10 marks**Q5** ✓**Q4 Digital Certificate Verification using Set Interface** 10 marks

Digital Certificate Verification using Set Interface

Store certificate IDs using an appropriate Set implementation.

Prevent duplicate certificate entries.

Verify whether a certificate ID exists.

Display certificates in insertion order.

Display certificates in sorted order.

Find the total number of unique certificates issued.

Remove revoked certificates.

Requirement: Use suitable Set implementations.

Your Code

```
1 import java.util.*;  
2 public class Employee {
```

Input6
C101

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using set interface
10 marks**Q5** ✓
Smart Inventory Management using Map Interface
10 marks

5/5 answered

Your Code

```
1 import java.util.*;  
2 public class Employee {  
3     public static void main(String[] args) {  
4         Scanner sc = new Scanner(System.in);  
5         LinkedHashSet<String> certificates = new LinkedHashSet<>();  
6         int n = sc.nextInt();  
7         sc.nextLine();  
8         for (int i = 0; i < n; i++) {  
9             certificates.add(sc.nextLine());  
10        }  
11        String searchId = sc.nextLine();  
12        if (certificates.contains(searchId)) {  
13            System.out.println("Certificate Found");  
14        } else {  
15            System.out.println("Certificate Not Found");  
16        }  
17        System.out.println("Certificates in Insertion Order:");  
18        for (String id : certificates) {  
19            System.out.println(id);  
20        }  
21        System.out.println("Certificates in Sorted Order:");  
22        TreeSet<String> sorted = new TreeSet<>(certificates);  
23        for (String id : sorted) {  
24            System.out.println(id);  
25        }  
26    }  
27 }
```

Input6
C101
C102
C103**Output**Certificate Found
Certificates in Insertion Order:
C101
C102
C103
C104

```
13         System.out.println("Certificate Found");
14     } else {
15         System.out.println("Certificate Not Found");
16     }
17     System.out.println("Certificates in Insertion Order:");
18     for (String id : certificates) {
19         System.out.println(id);
20     }
21     System.out.println("Certificates in Sorted Order:");
22     TreeSet<String> sorted = new TreeSet<>(certificates);
23     for (String id : sorted) {
24         System.out.println(id);
25     }
26     System.out.println("Total Unique Certificates: " + certificates.size());
27     String revoked = sc.nextLine();
28     certificates.remove(revoked);
29     System.out.println("Certificates After Revocation:");
30     for (String id : certificates) {
31         System.out.println(id);
32     }
33     sc.close();
34 }
35 }
```

 Run Save

Q5.



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Interface
10 marks

Q4
Digital Certificate Verification
using Set Interface
10 marks

Q5 Smart Inventory Management using Map Interface

10 marks

Smart Inventory Management using Map Interface

Store product information using an appropriate Map implementation.

Add new products.

Update product quantity.

Search a product using Product ID.

Calculate the total inventory value.

Display products category-wise.

Remove discontinued products.

Display all available products.

Requirement: Use a suitable Map implementation.

Your Code

Input



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Using Set Interface
10 marks

Q5
Smart Inventory Management
using Map Interface
10 marks

5/5 answered

Your Code

```
1 import java.util.*;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         HashMap<Integer, Product> map = new HashMap<>();
6         int n = sc.nextInt();
7         for (int i = 0; i < n; i++) {
8             int id = sc.nextInt();
9             sc.nextLine();
10            String name = sc.nextLine();
11            String category = sc.nextLine();
12            int quantity = sc.nextInt();
13            double price = sc.nextDouble();
14            map.put(id, new Product(id, name, category, quantity, price));
15        }
16        int searchId = sc.nextInt();
17        if (map.containsKey(searchId)) {
18            Product p = map.get(searchId);
19            System.out.println("Product Found");
20            System.out.println(p.id + " " + p.name + " " + p.category);
21        } else {
22            System.out.println("Product Not Found");
23        }
24    }
25 }
```

Input

```
5
50000
102
Mouse
Electronics
```

Output

```
Product Found
101 Laptop Electronics 5 50000.0
Total Inventory Value = 263000.0
101 Laptop 5 50000.0
102 Mouse 20 500.0
101 Laptop Electronics 5 50000.0
```



```

16     int searchId = sc.nextInt();
17     if (map.containsKey(searchId)) {
18         Product p = map.get(searchId);
19         System.out.println("Product Found");
20         System.out.println(p.id + " " + p.name + " " + p.category);
21     } else {
22         System.out.println("Product Not Found");
23     }
24     double total = 0;
25     for (Product p : map.values()) {
26         total += p.quantity * p.price;
27     }
28     System.out.println("Total Inventory Value = " + total);
29     sc.nextLine();
30     String cat = sc.nextLine();
31     for (Product p : map.values()) {
32         if (p.category.equalsIgnoreCase(cat)) {
33             System.out.println(p.id + " " + p.name + " " + p.quantity);
34         }
35     }
36     for (Product p : map.values()) {
37         System.out.println(p.id + " " + p.name + " " + p.category);
38     }
39 }
40 }
41 class Product {

```

```

38     }
39 }
40 }
41 class Product {
42     int id;
43     String name;
44     String category;
45     int quantity;
46     double price;
47     Product(int id, String name, String category, int quantity, double price) {
48         this.id = id;
49         this.name = name;
50         this.category = category;
51         this.quantity = quantity;
52         this.price = price;
53     }
54 }

```

 Run

 Save